

Day 2: Log Source Inventory Table

Document Title: Centralized Logging Architecture
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Objective

To establish a centralized log management architecture by configuring log forwarding from a Linux host to a central collector, deploying enhanced endpoint monitoring on Windows via Sysmon, and documenting the critical Windows Event IDs required for SIEM source onboarding.

System Information

Detail	Value
Log Sender (Linux)	UbuntuServer 22.04 — 192.168.56.104
Log Receiver/Collector	Kali Linux — 192.168.56.102
Endpoint Monitored	Windows 10 VM
Transport Protocol	TCP/UDP, Port 514
Sysmon Config Source	SwiftOnSecurity (GitHub)

Log Source Inventory Table

Source Name	Format	Transport	What It Detects
Linux auth.log	Syslog (plaintext)	rsyslog, TCP forward, port 514	SSH logins, sudo usage, authentication successes and failures
Linux syslog	Syslog (plain text)	rsyslog, TCP forward, port 514	General system activity, service starts/stops, daemon events
Linux kern.log	Syslog (plain text)	rsyslog, TCP forward, port 514	Kernel-level events, hardware/driver issues, low-level system errors
Windows Security Log	Windows binary/XML	Local Event Viewer	Logon/logoff events, account management, privilege use, audit policy changes

Critical Windows Event IDs Reference

Event ID	Source	What It Means	What Attack It Detects
4624	Security	Successful logon	Confirms who logged in and how — baseline for unusual login detection
4625	Security	Failed logon	Repeated occurrences indicate brute force password attacks
4672	Security	Special privileges assigned to logon	Detects privilege escalation to admin-level access
4688	Security	New process created	Core detection point for malware execution
4698	Security	Scheduled task created	Detects attacker persistence via scheduled tasks
4719	Security	Audit policy changed	Detects attacker attempting to disable logging
4720	Security	User account created	Detects backdoor account creation
7045	System	New service installed	Common malware persistence technique
1102	Security	Audit log cleared	Critical indicator of evidence destruction
Sysmon Event ID 1	Sysmon	Process creation (detailed)	Richer alternative to 4688 — full command line, parent process, file hash

Attack chain pattern these events reveal together:

4625 (failed logins) → 4624 (successful breach) → 4672 (privilege escalation) → 4688/Sysmon 1 (malicious execution) → 4720/7045 (persistence established) → 4719/1102 (evidence covered)

Practical Work Completed

Part 1 — Centralized Syslog Forwarding

Configured rsyslog on Ubuntu Server to forward authentication and system logs via TCP to a central Kali log collector on port 514. Verified end-to-end log delivery with test messages and confirmed real authentication events (SSH logins) were being forwarded automatically in real time.

Step 1

Installed (or verified the installation of) the **rsyslog** package to ensure the logging service was available on the system.

```
Session Actions Edit View Help
(fara@kali)-[~/Desktop]
$ sudo apt update
sudo apt install rsyslog -y
```

Step 2

Opened the `rsyslog.conf` file with Nano using root privileges to configure the rsyslog service.

```
(fara@kali)-[~]
$ sudo nano /etc/rsyslog.conf
[sudo] password for fara:
```

Step 3

Enabled UDP and TCP listeners on port **514** to allow the system to receive remote syslog messages

```
GNU nano 9.0 /etc/rsyslog.conf *
module(load="imudp")
input(type="imudp" port="514")
module(load="imtcp")
input(type="imtcp" port="514")
```

Step 4

Verified that rsyslog was listening on TCP and UDP port **514**.

```
(fara@kali)-[~]
$ sudo ss -tulnp | grep 514
udp    UNCONN 0      0      0.0.0.0:514
       0.0.0.0:*    users:(("rsyslogd",pid=1663
6,fd=6))
udp    UNCONN 0      0      [::]:514
       [::]:*      users:(("rsyslogd",pid=1663
6,fd=7))
tcp    LISTEN 0      25     0.0.0.0:514
       0.0.0.0:*    users:(("rsyslogd",pid=1663
6,fd=8))
tcp    LISTEN 0      25     [::]:514
       [::]:*      users:(("rsyslogd",pid=1663
6,fd=9))
```

Step 5

Configured the Kali log server to store incoming remote logs in separate directories based on the hostname.

```
GNU nano 9.0 /etc/rsyslog.conf *
#### RULES ####
#####

#
# Log anything besides private authentication messages to a single file
#
*. *;auth,authpriv.none                -/var/log/syslog

#
# Log commonly used facilities to their own log file
#
auth,authpriv.*                       /var/log/auth.log
cron.*                                -/var/log/cron.log
kern.*                                 -/var/log/kern.log
mail.*                                 -/var/log/mail.log
user.*                                 -/var/log/user.log

#
# Emergencies are sent to everybody logged in.
#
*.emerg                                :omusrmsg:*
# Store logs received from Ubuntu Server separately
$template RemoteLogs, "/var/log/remote/%HOSTNAME%/%PROGRAMNAME%.log"
*. * ?RemoteLogs
& stop
```

Step 6

Created the remote log storage directory, set the required permissions, and restarted rsyslog

```
(fara@kali)-[~/Desktop]
$ sudo mkdir -p /var/log/remote
$ sudo chown root:adm /var/log/remote

(fara@kali)-[~/Desktop]
$ sudo systemctl restart rsyslog
```

Step 7

Allowed TCP and UDP traffic on port **514** through the firewall for remote log collection.

```
(fara@kali)-[~/Desktop]
$ sudo ufw allow 514/tcp
$ sudo ufw allow 514/udp
Rule added
Rule added (v6)
Rule added
Rule added (v6)
```

Step 8

Created a new rsyslog forwarding configuration file on the Ubuntu Server.

```
Session Actions Edit View Help
fara@ubuntu-server:~$ sudo nano /etc/rsyslog.d/
50-forward.conf
```

Step 9

Configured the Ubuntu Server to forward authentication and system logs to the Kali log server.

```
Session Actions Edit View Help
/etc/rsyslog.d/50-forward.conf *
# Forward auth, syslog, and kern logs to cent>
auth,authpriv.* @192.168.56.102:514
*. * @192.168.56.102:514
```

Step 10

Restarted the rsyslog service on the Ubuntu Server to apply the forwarding configuration

```
fara@ubuntuuserver: ~$ sudo systemctl restart rsyslog
```

Step 11

Tested connectivity to the Kali server, generated a test log message, and verified that the forwarded logs were successfully received and stored on the Kali log server.

```
Session Actions Edit View Help
fara@ubuntuuserver:~$ nc -zv 192.168.56.102 514
Connection to 192.168.56.102 514 port [tcp/shell] succeeded!
fara@ubuntuuserver:~$ logger "TEST MESSAGE - Day 2 log forwarding check"
fara@ubuntuuserver:~$
```

```
(fara@kali)-[~/Desktop]
$ sudo cat /var/log/remote/ubuntuuserver/*.log
2026-06-30T14:39:40+05:30 ubuntuuserver fara: TEST MESSAGE - Day 2 log forwarding check
2026-06-30T14:36:44+05:30 ubuntuuserver rsyslogd: message repeated 6 times: [cannot connect to 192.168.56.102:514: No route to host [v8.2112.0 try https://www.rsyslog.com/e/2027 ]]
2026-06-30T14:36:44+05:30 ubuntuuserver rsyslogd: action 'action-8-builtin:omfwd' resumed (module 'builtin:omfwd') [v8.2112.0 try https://www.rsyslog.com/e/2359 ]
2026-06-30T14:36:44+05:30 ubuntuuserver rsyslogd: action 'action-9-builtin:omfwd' resumed (module 'builtin:omfwd') [v8.2112.0 try https://www.rsyslog.com/e/2359 ]
2026-06-30T14:36:44+05:30 ubuntuuserver sshd[2605]: Accepted password for fara from 192.168.56.102 port 58410 ssh2
2026-06-30T14:36:44+05:30 ubuntuuserver sshd[2605]: Accepted password for fara from 192.168.56.102 port 58410 ssh2
2026-06-30T14:36:44+05:30 ubuntuuserver sshd[2605]: pam_unix(sshd:session): session opened for user fara(uid=1000) by (uid=0)
2026-06-30T14:36:44+05:30 ubuntuuserver sshd[2605]: pam_unix(sshd:session): session opened for user fara(uid=1000) by (uid=0)
2026-06-30T14:36:44+05:30 ubuntuuserver systemd[1]: Started Session 12 of User fara.
2026-06-30T14:36:44+05:30 ubuntuuserver systemd-logind[759]: New session 12 of user fara.
2026-06-30T14:36:44+05:30 ubuntuuserver systemd-logind[759]: New session 12 of user fara.
```

Part 2 — Sysmon Deployment on Windows

Installed Sysmon on a Windows 10 VM using the SwiftOnSecurity configuration — an industry-standard, pre-tuned ruleset widely used in production SOC environments. Verified the Sysmon64 service was running and confirmed live event generation in Event Viewer, observing Process Creation (Event ID 1) and DNS Query (Event ID 22) events populating in real time.

Step 1

Installed **Sysmon** using the selected XML configuration file to enable advanced system activity monitoring.

```
Administrator: Command Prompt
C:\Windows\system32>cd c:\Users\farhan\Downloads\Sysmon
c:\Users\farhan\Downloads\Sysmon>Sysmon64.exe -i sysmonconfig-export.xml

System Monitor v15.21 - System activity monitor
By Mark Russinovich and Thomas Garnier
Copyright (C) 2014-2026 Microsoft Corporation
Using libxml2. libxml2 is Copyright (C) 1998-2012 Daniel Veillard. All Rights Reserved.
Sysinternals - www.sysinternals.com

Loading configuration file with schema version 4.50
Sysmon schema version: 4.91
Configuration file validated.
Sysmon64 installed.
SysmonDrv installed.
Starting SysmonDrv.
SysmonDrv started.
Starting Sysmon64..
Sysmon64 started.

c:\Users\farhan\Downloads\Sysmon>_
```

Step 2

Verified that the **Sysmon** service was installed successfully and running using the `sc query` command.

```
c:\Users\farhan\Downloads\Sysmon>sc query Sysmon64

SERVICE_NAME: Sysmon64
        TYPE               : 10  WIN32_OWN_PROCESS
        STATE                : 4   RUNNING
                                (STOPPABLE, NOT_PAUSABLE, IGNORES_SHUTDOWN)
        WIN32_EXIT_CODE       : 0   (0x0)
        SERVICE_EXIT_CODE   : 0   (0x0)
        CHECKPOINT           : 0x0
        WAIT_HINT            : 0x0
```


Step 3

Opened the **Sysmon Operational** log in Event Viewer to confirm that Sysmon was generating and recording events.

Operational Number of events: 151 (!) New events available					
Level	Date and Time	Source	Event ID	Task Ca...	
Information	30/06/2026 12:19:42 PM	Sysmon	5	Proces...	
Information	30/06/2026 12:19:36 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:21 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:21 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:20 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:19 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:18 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:16 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:15 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:14 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:01 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:19:00 PM	Sysmon	22	Dns qu...	
Information	30/06/2026 12:19:00 PM	Sysmon	22	Dns qu...	
Information	30/06/2026 12:18:59 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:18:59 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:18:59 PM	Sysmon	1	Proces...	
Information	30/06/2026 12:18:54 PM	Sysmon	22	Dns qu...	
Event 22, Sysmon					

Step 4

Viewed a **Process Creation (Event ID 1)** log to examine the detailed information captured by Sysmon for a newly created process.

Event Properties - Event 1, Sysmon

GeneralDetails

Process Create:
RuleName: -
UtcTime: 2026-06-30 19:33:42.763
ProcessGuid: {d0134298-1a16-6a44-5404-000000000e00}
ProcessId: 6944
Image: C:\Windows\System32\notepad.exe
FileVersion: 10.0.19041.5794 (WinBuild.160101.0800)

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 1
Level: Information
User: SYSTEM
OpCode: Info
More Information: [Event Log Online Help](#)

Logged: 30/06/2026 12:33:42 PM
Task Category: Process Create (rule: ProcessCreati
Keywords:
Computer: farhansoc.soc.lab

Conclusion

Day 2 established the foundational log management architecture required for SIEM integration. A working centralized syslog pipeline was built and verified between Ubuntu Server and Kali, demonstrating real-time forwarding of authentication and system events. Sysmon was deployed on Windows with a production-grade configuration, providing significantly enhanced visibility beyond default Windows logging. The documented inventory of log sources and critical Event IDs forms the baseline reference that will support Wazuh and Splunk source onboarding in upcoming tasks.